Concentration Inequalities (AI2200)

Lecture 05

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Recap: Bernoulli Averages Concentration

- Suppose that $X_1, X_2, \ldots \overset{\text{IID}}{\sim} \mathrm{Ber}(p)$ for some fixed $p \in (0,1)$
- Then, for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left\{\left|\frac{1}{n}\sum_{k=1}^{n} X_k - p\right| \geq \varepsilon p\right\}\right) \leq 2\,\exp\left(-\frac{np\varepsilon^2}{2+\varepsilon}\right).$$

- Given a feature space $\mathcal{X}$, a finite label space $\mathcal{Y}$, and a finite collection $\mathcal{G}$ of classifiers mapping $\mathcal{X}$ to $\mathcal{Y}$,

$$\mathbb{P}\left(\left\{R(\widehat{g}_n) \geq R(g^\star) + 2\varepsilon\right\}\right) \leq 2|\mathcal{G}|\,\exp\left(-\frac{n\varepsilon^2 R^2(g^\star)}{2+\varepsilon}\right) \qquad \forall\, \varepsilon > 0.$$

Here:

$$R(g) := \mathbb{P}_{(X,Y)\sim P_{X,Y}}\left(\{g(X) \neq Y\}\right), \qquad g \in \mathcal{G},$$

$$\widehat{g}_n := \arg\min_{g\in\mathcal{G}} \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{g(X_k) \neq Y_k\}},$$

$$g^\star := \arg\min_{g\in\mathcal{G}} R(g).$$

Gaussian Concentration

- Beyond Bernoulli, what can we say about the concentration phenomena for other distributions?
- Suppose that $X \sim \mathcal{N}(\mu, \sigma^2)$. Then,

Exercise:
$$M_X(t) \;=\; \exp\left(\mu t + \frac{t^2\sigma^2}{2}\right), \qquad t \in \mathbb{R}.$$

In this case, we have

Exercise:
$$\mathbb{P}\Big(\{X - \mu \geq \varepsilon\}\Big) \;\leq\; \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right) \qquad \forall\, \varepsilon > 0,$$

Exercise:
$$\mathbb{P}\Big(\{X - \mu \leq -\varepsilon\}\Big) \;\leq\; \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right) \qquad \forall\, \varepsilon > 0.$$

- As a consequence, it follows that if $X_1, \ldots, X_n \overset{\text{IID}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{P}\left(\left\{\left|\frac{1}{n}\sum_{k=1}^{n} X_k \;-\; \mu\right| \geq \varepsilon\right\}\right) \;\leq\; 2\,\exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right) \qquad \forall\, \varepsilon > 0.$$

Rademacher Concentration

- Suppose that $X_1, \ldots, X_n$ are IID and **Rademacher** distributed, i.e.,

$$\mathbb{P}\Big(\{X_1 = +1\}\Big) = \frac{1}{2} = \mathbb{P}\Big(\{X_1 = -1\}\Big).$$

- Given fixed constants $a_1, \ldots, a_n \in \mathbb{R}$, let

$$Y := \sum_{k=1}^{n} a_k X_k; \qquad \mathbb{E}[Y] = 0; \qquad \mathrm{Var}(Y) = \sum_{k=1}^{n} a_k^2 = \|a\|^2.$$

- For any $\varepsilon > 0$, we have

$$\mathbb{P}\left(\left\{\sum_{k=1}^{n} a_k X_k \geq \varepsilon\right\}\right) \leq \inf_{t>0} \frac{\mathbb{E}\left[\exp\left(t\sum_{k=1}^{n} a_k X_k\right)\right]}{\exp(t\varepsilon)} = \inf_{t>0} \frac{\prod_{k=1}^{n} \mathbb{E}[\exp(ta_k X_k)]}{\exp(t\varepsilon)} = \inf_{t>0} \frac{\prod_{k=1}^{n} \left(\frac{e^{ta_k} + e^{-ta_k}}{2}\right)}{e^{t\varepsilon}}$$

$$\overset{(a)}{\leq} \inf_{t>0} \frac{\prod_{k=1}^{n} e^{t^2 a_k^2/2}}{e^{t\varepsilon}} = \exp\left(-\frac{\varepsilon^2}{2\|a\|^2}\right) \qquad \left((a): \frac{e^x + e^{-x}}{2} \leq e^{x^2/2}, \quad x \in \mathbb{R}\right)$$

Sub-Gaussian Random Variables

Sub-Gaussian Random Variables

Definition (Sub-Gaussian Random Variable)

A random variable X is said to be **sub-Gaussian** if there exists a constant $K > 0$ such that

$$\mathbb{P}\Big(\{|X| \geq \varepsilon\}\Big) \;\leq\; 2\,\exp\left(-\frac{\varepsilon^2}{K^2}\right) \qquad \forall\,\varepsilon > 0.$$

Examples:

- If $X \sim \mathcal{N}(0,1)$, then X is sub-Gaussian
- Any bounded random variable (e.g., Bernoulli, Binomial, Beta, Uniform) is sub-Gaussian
- **A random variable that is NOT sub-Gaussian:** $X \sim \mathrm{Exponential}(\lambda)$ for a fixed $\lambda > 0$

Sub-Gaussian Random Variables

Theorem 1 (Equivalent Statements for sub-Gaussian Random Variables)

The following statements are equivalent.

1. *There exists a constant* $K_1 > 0$ *such that* $\mathbb{P}\left(\{|X| \geq \varepsilon\}\right) \leq 2\, e^{-\varepsilon^2/K_1^2} \quad \forall\, \varepsilon > 0.$
2. *There exists a constant* $K_2 > 0$ *such that* $\left(\mathbb{E}\left[|X|^p\right]\right)^{1/p} \leq K_2 \sqrt{p} \quad \forall\, p \geq 1.$
3. *There exists a constant* $K_3 > 0$ *such that* $\mathbb{E}\left[e^{t^2X^2}\right] \leq e^{t^2K_3^2} \quad \forall\, t \in \left[-\frac{1}{K_3}, \frac{1}{K_3}\right].$
4. *There exists a constant* $K_4 > 0$ *such that* $\mathbb{E}\left[e^{X^2/K_4^2}\right] \leq 2.$
5. **(Applicable if $\mathbb{E}[X] = 0$)**
   There exists a constant $K_5 > 0$ *such that* $\mathbb{E}\left[e^{tX}\right] \leq e^{K_5^2t^2} \quad \forall\, t \in \mathbb{R}.$

Notes:

- If any of the above statements holds for some constant K, it also holds for a larger constant than K
- Any pair of constants in the set $\{K_1, \ldots, K_5\}$ satisfies the relation

$$K_i/K_j = c_{i,j},$$

where $c_{i,j}$ is a universal constant that does not depend on the distribution of X!

Sub-Gaussian Norm

Definition (Sub-Gaussian Norm)

For a sub-Gaussian random variable X, its **sub-Gaussian norm** is defined as

$$\|X\|_{\psi_2} := \inf\left\{K > 0 : \mathbb{E}\left[e^{X^2/K^2}\right] \le 2\right\}.$$

- If X is sub-Gaussian, then αX is sub-Gaussian for all $\alpha \in \mathbb{R}$. Furthermore,

$$\|\alpha X\|_{\psi_2} = |\alpha| \, \|X\|_{\psi_2}.$$

Proof:

$$\|\alpha X\|_{\psi_2} = \inf\left\{K > 0 : \mathbb{E}\left[e^{\alpha^2 X^2/K^2}\right] \le 2\right\} = \inf\left\{K > 0 : \mathbb{E}\left[e^{X^2/(K/|\alpha|)^2}\right] \le 2\right\}$$

$$\overset{K=M|\alpha|}{=} |\alpha| \cdot \inf\left\{M > 0 : \mathbb{E}\left[e^{X^2/M^2}\right] \le 2\right\} = |\alpha| \, \|X\|_{\psi_2}.$$

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- If X, Y are sub-Gaussian with sub-Gaussian norms $K_X = \|X\|_{\psi_2}$ and $K_Y = \|Y\|_{\psi_2}$ respectively, then

$$\|X+Y\|_{\psi_2} \;\leq\; \|X\|_{\psi_2} + \|Y\|_{\psi_2} \;=\; K_X + K_Y.$$

Proof:

- It suffices to prove that $\mathbb{E}\left[e^{(X+Y)^2/(K_X+K_Y)^2}\right] \;\leq\; 2.$
- Let $g(x) = e^{x^2}$. Noting that g is increasing and convex, we get

$$\begin{aligned}
\mathbb{E}\left[e^{(X+Y)^2/(K_X+K_Y)^2}\right] &= \mathbb{E}\left[g\left(\frac{|X+Y|}{K_X+K_Y}\right)\right] \;\leq\; \mathbb{E}\left[g\left(\frac{|X+|Y|}{K_X+K_Y}\right)\right] \\
&= \mathbb{E}\left[g\left(\frac{K_X}{K_X+K_Y}\frac{|X|}{K_X} + \frac{K_Y}{K_X+K_Y}\frac{|Y|}{K_Y}\right)\right] \;\leq\; \frac{K_X}{K_X+K_Y}\,\mathbb{E}[e^{X^2/K_X^2}] + \frac{K_Y}{K_X+K_Y}\,\mathbb{E}[e^{Y^2/K_Y^2}] \\
&\leq\; 2.
\end{aligned}$$